“Heart Disease Prediction Web Application”

A PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this project report "**Heart Disease Prediction Web Application**" is the Bonafide work of PRANAV SRIRAM (230701235), PRANESH SENTHILKUMAR (230701236), PRATHYUSH RK who carried out the project work under my supervision.

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ABSTRACT

The Heart Disease Prediction System is an intelligent web-based healthcare application developed to assist in the early detection and risk assessment of cardiovascular diseases using Machine Learning techniques. The primary objective of this project is to provide users and healthcare professionals with a reliable and data-driven platform capable of predicting the likelihood of heart disease based on a patient's clinical and medical attributes. Early diagnosis of heart disease is critical for reducing mortality rates and improving treatment outcomes; therefore, this system aims to support preventive healthcare by offering quick and accurate predictions through an automated digital solution.

The application is built using a full-stack architecture that combines a Flask-based backend with an interactive frontend developed using HTML5, CSS3, and Flask Templates. The backend is responsible for handling API requests, processing user input, integrating machine learning models, and generating prediction results. SQLite3 is used as the database for managing user authentication and securely storing patient-related records. Core Python libraries such as NumPy and Pandas are utilized for efficient data preprocessing and analysis, while Pickle is employed for loading pre-trained machine learning models into the application.

The prediction mechanism is based on an ensemble learning approach that integrates five different Machine Learning algorithms: Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Decision Tree Classifier, and Random Forest Classifier. Instead of depending on a single model, the system combines the outputs of all five algorithms to improve prediction reliability and reduce bias. Each model independently analyzes the patient's medical parameters, including age, gender, blood pressure, cholesterol levels, fasting blood sugar, ECG results, chest pain type, maximum heart rate, exercise-induced angina, and other clinical indicators. The final prediction is determined through a majority voting mechanism, where the consensus among the models decides whether the patient is at low or high risk of heart disease. Additionally, the system calculates a confidence score based on the level of agreement among the models.

The Heart Disease Prediction System provides users with an intuitive interface for entering medical data and viewing prediction results in real time. By combining Artificial Intelligence, Machine Learning, and web technologies, the project demonstrates how modern predictive systems can support healthcare professionals in clinical decision-making and help individuals become more aware of their cardiovascular health. The system serves as a scalable foundation for future healthcare applications involving predictive analytics, medical decision support, and intelligent health monitoring solutions.

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CHAPTER 1

INTRODUCTION

Heart disease is one of the leading causes of death worldwide, affecting millions of people every year. Early detection and proper diagnosis play a crucial role in preventing severe cardiovascular complications and improving patient survival rates. However, traditional diagnosis methods often require multiple medical tests, expert consultation, and considerable time for analysis. With the rapid advancement of Artificial Intelligence and Machine Learning technologies, healthcare systems are increasingly adopting intelligent solutions to assist doctors and patients in predicting diseases more efficiently and accurately.

The Heart Disease Prediction System is a web-based healthcare application developed to predict the likelihood of heart disease using Machine Learning algorithms. The system analyzes various clinical parameters such as age, blood pressure, cholesterol level, chest pain type, ECG results, heart rate, and other medical indicators to determine whether a patient is at risk of developing heart disease. The project aims to provide a fast, reliable, and user-friendly diagnostic support tool that can help in early detection and preventive healthcare.

The application is developed using Flask as the backend framework, HTML5 and CSS3 for the frontend interface, and SQLite3 for database management. Machine Learning models are implemented using Scikit-learn, while NumPy and Pandas are used for data preprocessing and analysis. The system adopts an ensemble learning approach by integrating multiple algorithms including Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Decision Tree, and Random Forest. The final prediction is generated using a majority voting mechanism among these models to improve prediction accuracy and reliability.

The proposed system not only predicts the risk level but also provides confidence scores based on model agreement. This helps users understand the reliability of the prediction and supports healthcare professionals in making informed decisions. The project demonstrates how Machine Learning and web technologies can be effectively combined to create intelligent healthcare applications capable of supporting modern medical practices.

CHAPTER 2

LITERATURE SURVEY

Machine Learning (ML) has significantly transformed the healthcare industry by enabling intelligent systems capable of assisting in disease diagnosis, prediction, and clinical decision-making. Among various healthcare applications, heart disease prediction has gained considerable attention due to the increasing number of cardiovascular-related deaths worldwide. Researchers have explored multiple Machine Learning techniques to improve prediction accuracy, automate diagnosis, and support early detection of heart-related illnesses. This section presents an overview of existing research and technologies related to heart disease prediction systems. It highlights how different ML concepts, including classification algorithms, ensemble learning, probabilistic reasoning, and intelligent healthcare systems, are applied in predictive medical applications.

An intelligent healthcare prediction system functions as an intelligent agent by collecting patient information, analyzing clinical parameters, and generating diagnostic predictions. In Paper [1], Russell and Norvig introduced the concept of intelligent agents capable of perceiving environmental inputs and taking rational actions based on reasoning mechanisms. Extending this concept, Paper [2] proposed a clinical decision-support system that acts as a rational healthcare agent by analyzing patient symptoms and medical history to assist doctors in diagnosis. The Heart Disease Prediction System follows a similar approach where the application perceives user-provided health data, reasons using Machine Learning algorithms, and acts by generating prediction results along with confidence scores.

Classification algorithms play a major role in medical diagnosis systems. Researchers have implemented several supervised Machine Learning techniques for predicting cardiovascular diseases. Paper [3] discussed the use of Logistic Regression for heart disease prediction because of its simplicity and effectiveness in binary classification problems. The study showed that Logistic Regression provides reliable results when analyzing structured clinical datasets. Similarly, Paper [4] applied K-Nearest Neighbors (KNN) for classifying patients into different risk categories based on similarity between patient records. The model achieved good prediction accuracy for smaller datasets and demonstrated effectiveness in identifying high-risk patients.

Support Vector Machine (SVM) algorithms are widely used in healthcare applications because of their ability to handle complex and high-dimensional medical data. Paper [5] proposed an SVM-based heart disease prediction system that utilized kernel functions to improve classification accuracy. The results showed that SVM performed efficiently in separating healthy and diseased patient classes. Decision Tree algorithms have also been applied in medical analytics due to their interpretability and rule-based structure. In Paper [6], researchers developed a Decision Tree-based prediction system that generated understandable diagnostic rules for doctors and healthcare professionals. Random Forest, an advanced ensemble learning technique, was later introduced to improve the limitations of single Decision Trees by reducing overfitting and improving generalization.

Ensemble learning has become an important research area in disease prediction systems because combining multiple models often improves prediction reliability and accuracy. Paper [7] proposed an ensemble-based healthcare prediction model that integrated Logistic Regression, SVM, and Random Forest classifiers using majority voting techniques. The study reported higher accuracy compared to standalone models. Similarly, Paper [8] demonstrated that combining multiple Machine Learning algorithms reduces bias and increases robustness in medical predictions. The Heart Disease Prediction System follows a similar ensemble learning approach by integrating five Machine Learning models: Logistic Regression, K-Nearest Neighbors, Support Vector Machine, Decision Tree, and Random Forest. Each model independently predicts heart disease risk, and the final output is generated using a majority consensus mechanism.

Feature selection and preprocessing techniques are critical in healthcare prediction systems because medical datasets often contain missing values, redundant attributes, and noisy information. Paper [9] discussed the importance of preprocessing methods such as normalization, missing value handling, and feature scaling in improving Machine Learning model performance. These preprocessing techniques help algorithms generate more accurate and consistent predictions. Recent advancements in Explainable Artificial Intelligence (XAI) have also influenced healthcare applications by making prediction systems more transparent and understandable for users and doctors. The proposed Heart Disease Prediction System similarly performs preprocessing operations before feeding patient data into the trained models and provides reliable predictions through an intelligent ensemble-based approach.

Overall, the reviewed literature demonstrates that Machine Learning techniques have significantly improved heart disease prediction and clinical decision-making systems. Existing studies show that ensemble learning, intelligent classification algorithms, and probabilistic reasoning methods provide accurate and reliable prediction results. The proposed Heart Disease Prediction System builds upon these research contributions by integrating multiple Machine Learning algorithms into a user-friendly web application capable of providing real-time cardiovascular risk assessment and predictive healthcare support.

CHAPTER 3
SYSTEM
DESIGN

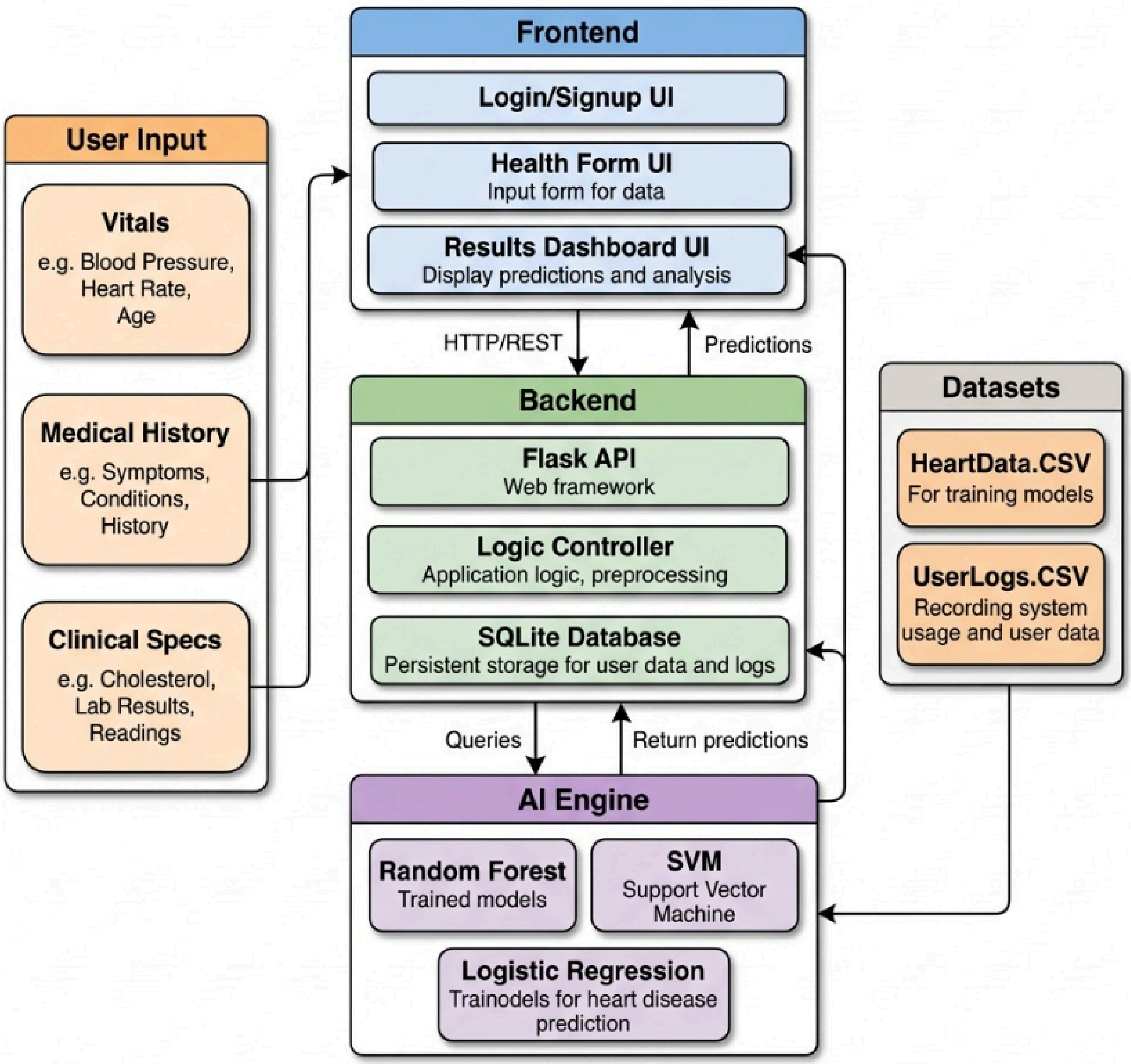
The proposed system, Heart Disease Prediction System, aims to provide an intelligent and automated solution for predicting the likelihood of heart disease using Machine Learning techniques. The system is designed to assist users and healthcare professionals by analyzing clinical and medical parameters to generate accurate risk predictions. Unlike traditional diagnosis methods that depend heavily on manual interpretation and multiple medical examinations, this system offers a fast, data-driven, and accessible platform capable of providing instant preliminary health assessments. The application improves early detection of cardiovascular diseases and supports preventive healthcare through intelligent prediction models.

The architecture of the proposed system follows a modular and scalable design approach. The frontend is developed using HTML5, CSS3, and Flask Templates, providing a responsive and user-friendly interface for patient data entry and result visualization. The backend is implemented using Flask (Python), which manages server-side operations, API routing, database connectivity, and communication with Machine Learning models. SQLite3 is used as the database for securely storing user authentication details and patient-related records. The modular architecture improves maintainability, scalability, and future extensibility of the system.

The core Machine Learning module performs prediction tasks using an ensemble learning approach. The system integrates five Machine Learning algorithms: Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Decision Tree Classifier, and Random Forest Classifier. Each algorithm independently analyzes patient clinical data such as age, blood pressure, cholesterol levels, chest pain type, ECG results, fasting blood sugar, heart rate, exercise-induced angina, and other cardiovascular indicators. The final prediction is generated through a majority voting mechanism, where the combined output of all models determines whether the patient is at low or high risk of heart disease. Data preprocessing techniques such as normalization, missing value handling, and feature formatting are also applied before prediction to improve model performance and accuracy.

The final output of the system includes a heart disease risk prediction along with confidence analysis and result visualization. By combining Machine Learning, data processing, and web technologies, the Heart Disease Prediction System provides an intelligent and efficient approach to predictive healthcare. The system also serves as a scalable foundation for future healthcare applications involving real-time monitoring, wearable device integration, and advanced AI-assisted diagnostic systems.

3.1 ARCHITECTURE DIAGRAM



System Overview:

The Heart Disease Prediction System accepts multiple clinical and medical inputs from users, including Age, Gender, Chest Pain Type, Blood Pressure, Cholesterol Level, ECG Results, Maximum Heart Rate, Fasting Blood Sugar, Exercise-Induced Angina, and other cardiovascular indicators, to generate heart disease risk predictions through an interconnected frontend-backend architecture.

Frontend Components:

The frontend layer consists of three main user interfaces:

- User Authentication UI: Allows users to register, log in, and securely access the application while maintaining patient privacy and record management.
- Health Input UI : Provides an interactive form where users can enter medical and clinical parameters required for heart disease prediction.
- Result Visualization UI : Displays prediction results, confidence scores, and health risk analysis in a simple and user-friendly format.

Backend Architecture:

The backend comprises three critical modules:

- Flask Application Modules: Contain the core application functionality, routing operations, and integration with Machine Learning models.
- Logic + Controller: Manages request processing, input validation, prediction handling, and communication between frontend, database, and ML models.
- Database: SQLite3 database stores user credentials, patient medical inputs, prediction history, and system records with secure bidirectional communication across backend components.

ML Layer:

The system employs five Machine Learning algorithms for intelligent prediction and analysis:

- Logistic Regression : Performs binary classification to predict whether the patient is at risk of heart disease.
- K-Nearest Neighbors (KNN) : Analyzes similarity between patient records to classify risk levels.
- Support Vector Machine (SVM) : Processes high-dimensional medical data and improves classification accuracy.
- Decision Tree : Generates rule-based predictions and provides interpretable decision-making structures.
- Random Forest : Combines multiple decision trees to improve prediction accuracy and reduce overfitting.
- Ensemble Voting Mechanism : All five models independently generate predictions, and the final output is determined using majority voting to improve reliability and consistency.

External Resources:

The system utilizes pre-trained Machine Learning models and medical datasets to improve prediction accuracy and healthcare analysis. External healthcare research datasets and Scikit-learn libraries support model training, evaluation, and prediction generation. The integration of Python libraries such as NumPy and Pandas enables efficient data processing, preprocessing, and feature analysis for reliable heart disease prediction.

Data Layer:

Four datasets and storage components power the system's prediction process:

- HeartDiseaseDataset.CSV : Contains patient clinical records, medical attributes, and heart disease prediction labels used for training and testing the ML models.
- UserData.CSV : Stores user-entered medical information and previous prediction records for analysis and history tracking.
- ProcessedData.CSV : Contains normalized and preprocessed patient data prepared for Machine Learning model input.
- ModelAccuracy.CSV : Stores accuracy scores, model evaluation metrics, and ensemble prediction performance details.

Data Flow:

The architecture follows this operational sequence: User medical inputs flow from the frontend interfaces to backend modules, where the Flask application validates and preprocesses the data. The backend then accesses the CSV datasets and passes the processed information to the Machine Learning layer. The five ML algorithms independently analyze the patient data and generate prediction results. The ensemble voting mechanism combines these outputs to determine the final heart disease risk prediction and confidence score. The results are stored in the database and returned through the backend to the frontend interfaces, where users can view prediction outcomes and health analysis reports through the interactive result dashboard.

CHAPTER 4

MODULES DESCRIPTION

4.1 MODULE 1

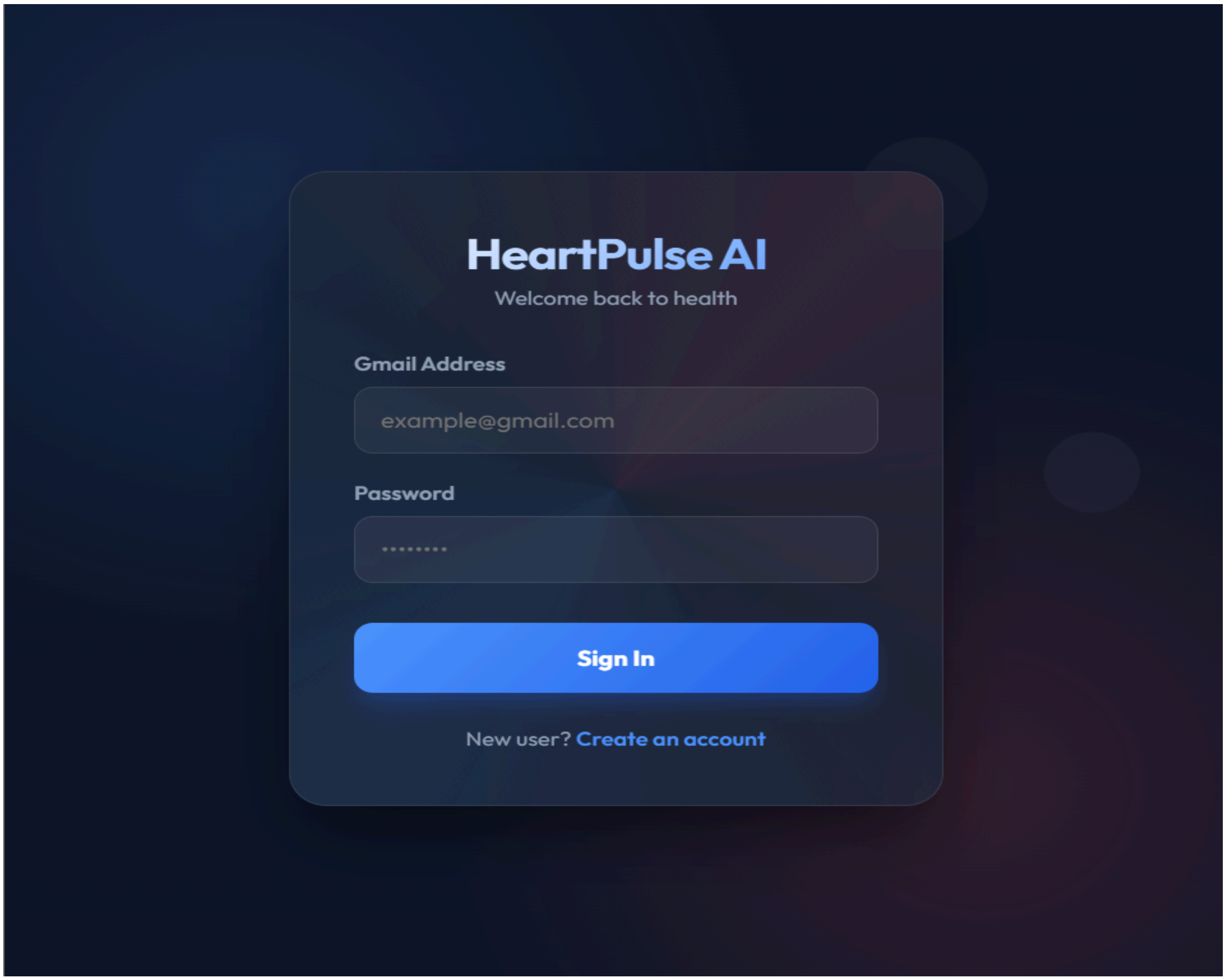


Figure 4.1 User Login and Dashboard

This page represents the main dashboard of the Heart Disease Prediction System, which serves as an intelligent healthcare assistant for predicting cardiovascular disease risks using Machine Learning algorithms. The application combines predictive analytics, medical data processing, and interactive visualization to provide users with a comprehensive healthcare monitoring experience.

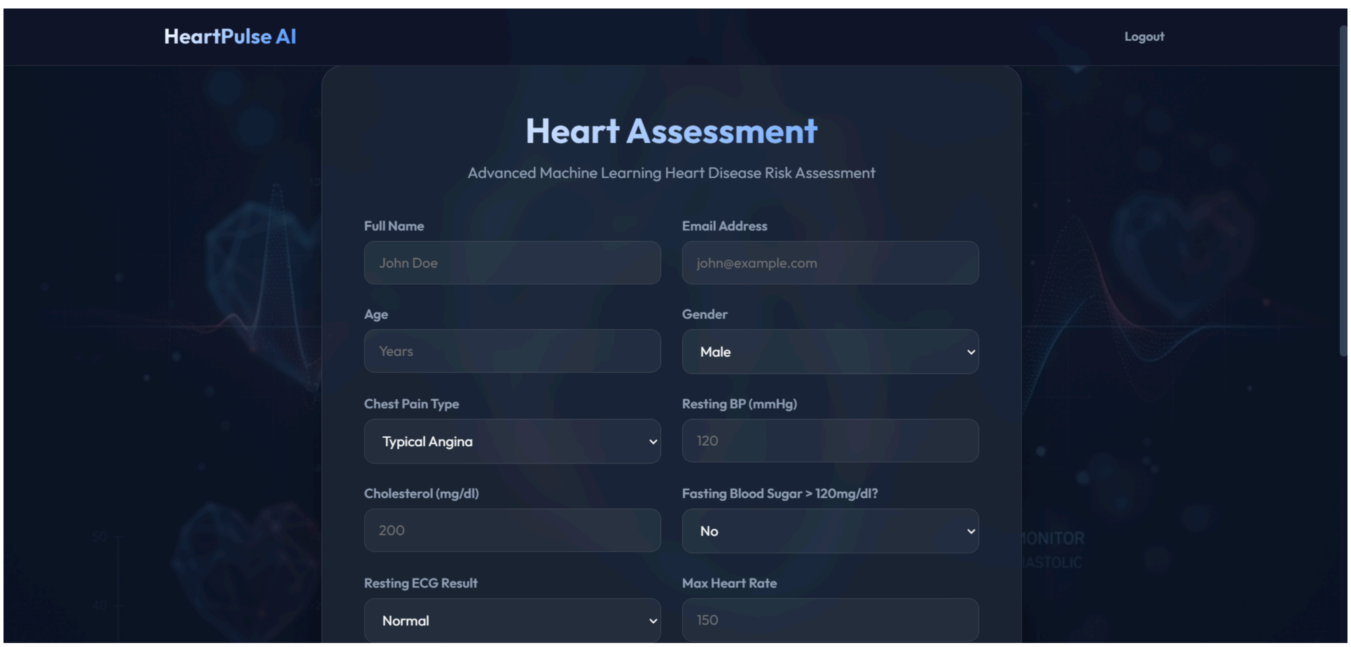
User Authentication and Profile Management :

Upon accessing the dashboard, users can securely register and log into the system using their credentials. The authentication module ensures privacy and secure handling of patient records through database-driven session management. Once logged in, users are welcomed with a personalized dashboard displaying their profile information and previous prediction history. This allows users to track their health assessments and monitor prediction results over time.

Health Data Input Interface:

The central component of this page is the medical data entry form, which collects important clinical parameters required for heart disease prediction. Users enter details such as age, gender, chest pain type, blood pressure, cholesterol level, fasting blood sugar, ECG results, maximum heart rate, exercise-induced angina, and other cardiovascular indicators. These inputs enable the system to generate accurate and personalized heart disease risk predictions using Machine Learning algorithms.

4.2 MODULE 2



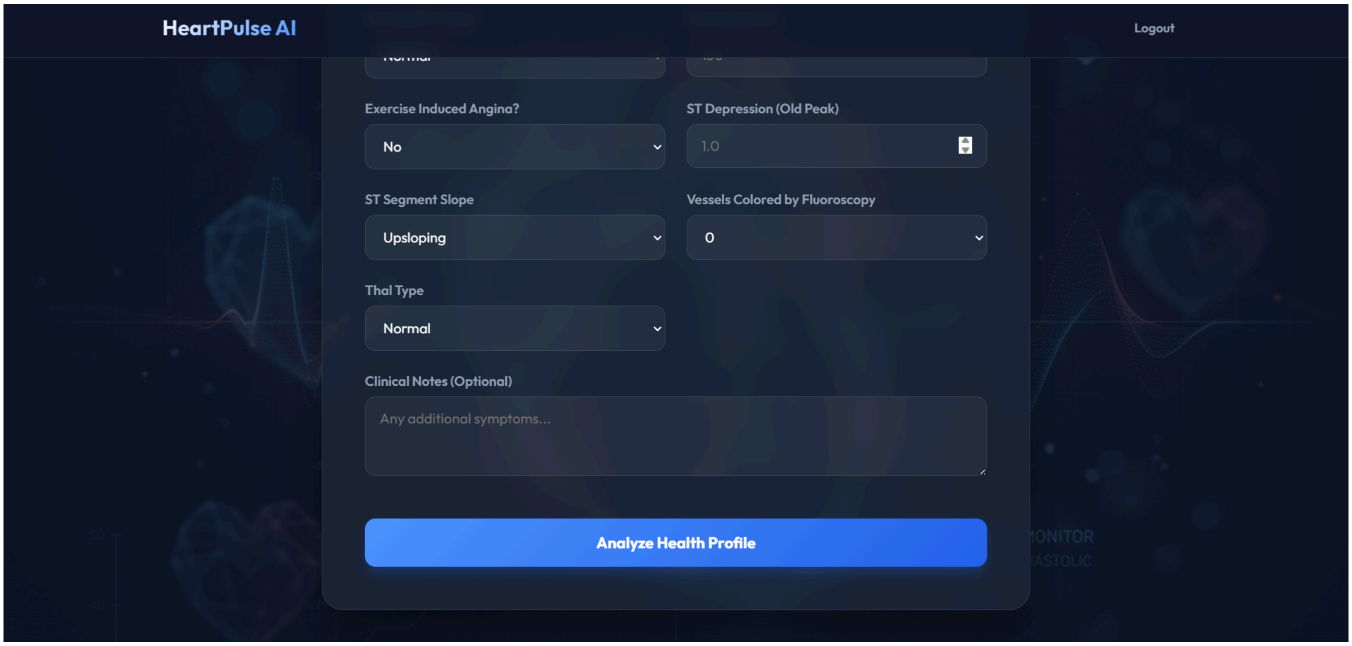


Figure 4.2 Health Assessment Dashboard

This page represents the main dashboard of the Heart Disease Prediction System, which serves as an intelligent healthcare assistant for predicting cardiovascular disease risk using Machine Learning algorithms. The application combines medical data analysis, ensemble Machine Learning models, and clinical report generation to provide a comprehensive healthcare prediction experience.

User Dashboard and Medical Input :

Upon accessing the dashboard, users are provided with a structured medical assessment form where they can enter important clinical parameters required for heart disease prediction. The interface is designed with a modern healthcare-themed layout featuring responsive cards, clean typography, and organized form fields to ensure ease of use and accessibility.

The dashboard collects multiple medical attributes including age, gender, chest pain type, resting blood pressure, cholesterol level, fasting blood sugar, ECG results, maximum heart rate, exercise-induced angina, ST depression, slope of ST segment, number of major vessels, and thalassemia test results. Users can also add optional clinical notes regarding symptoms or previous medical conditions.

Health Analysis Interface :

The primary component of this page is the “Analyze Health Profile” feature, which processes the entered medical data through multiple Machine Learning algorithms. Once the user submits the information, the backend validates and preprocesses the data before forwarding it to the prediction engine. This interface enables users to quickly receive personalized heart disease risk assessments based on clinical indicators and predictive analytics.

4.3 MODULE 3

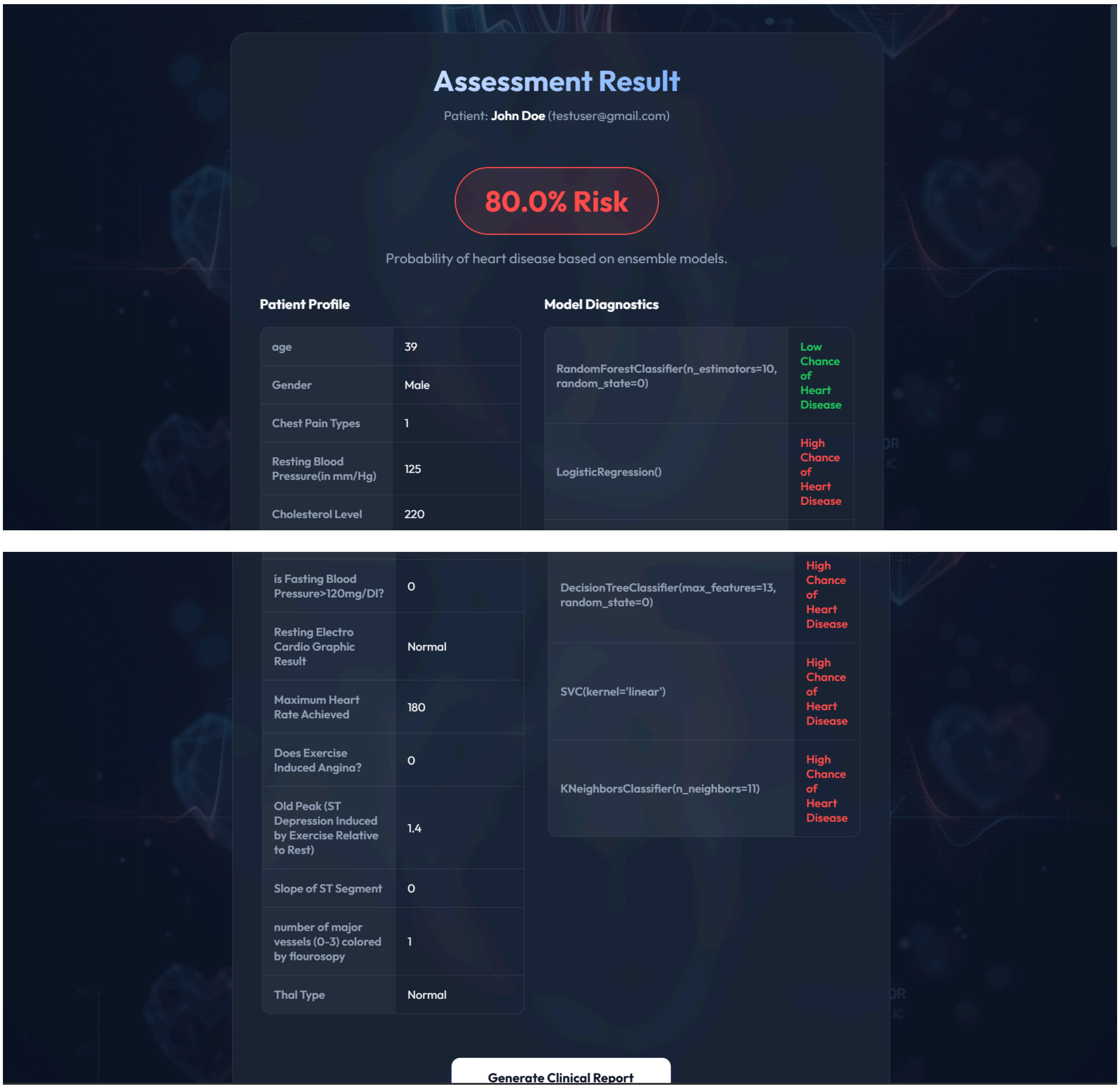


Figure 4.3 Assessment Result Dashboard

This page represents the Prediction Result Dashboard, which displays the complete heart disease analysis after users submit their medical information. It serves as a comprehensive diagnostic interface that presents patient details, prediction results, confidence scores, and Machine Learning model outputs in an organized and interactive format.

Risk Prediction and Confidence Analysis :

The dashboard prominently displays the predicted heart disease risk percentage generated using ensemble learning techniques. The result is highlighted using visual indicators and color-coded elements to clearly differentiate between low-risk and high-risk predictions. A confidence score is calculated based on the agreement among the Machine Learning models, improving transparency and reliability.

Patient Profile and Medical Parameters :

A dedicated section displays all patient clinical attributes submitted during the assessment process. Parameters such as blood pressure, cholesterol level, ECG results, heart rate, exercise-induced angina, and ST depression are organized in tabular form for easy readability and clinical reference.

Machine Learning Model Diagnostics :

The system uses five Machine Learning algorithms including Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Decision Tree, and Random Forest classifiers. Each model independently predicts the patient's condition, and the results are displayed alongside the final prediction. This helps users understand how different models contributed to the overall diagnosis.

The ensemble voting mechanism determines the final prediction by selecting the majority output among all algorithms. This approach improves prediction accuracy, minimizes bias, and enhances system robustness.

4.4 MODULE 4


5/15/26, 7:34 PM Analysis Result | HeartPulse AI

Patient: John Doe (testuser@gmail.com)

80.0% Risk

Probability of heart disease based on ensemble models.

Patient Profile		Model Diagnostics	
age	39	RandomForestClassifier(n_estimators=10, random_state=0)	Low Chance of Heart Disease
Gender	Male		
Chest Pain Types	1		
Resting Blood Pressure(in mm/Hg)	125	LogisticRegression()	High Chance of Heart Disease
Cholesterol Level	220		
is Fasting Blood Pressure>120mg/Dl?	0	DecisionTreeClassifier(max_features=13, random_state=0)	High Chance of Heart Disease
Resting Electro Cardio Graphic Result	Normal		
Maximum Heart Rate Achieved	180	SVC(kernel='linear')	High Chance of Heart Disease

127.0.0.1:5000/api 1/2

5/15/26, 7:34 PM Analysis Result | HeartPulse AI

Does Exercise Induced Angina?	0	KNeighborsClassifier(n_neighbors=11)	High Chance of Heart Disease
Old Peak (ST Depression Induced by Exercise Relative to Rest)	1.4		
Slope of ST Segment	0		
number of major vessels (0-3) colored by flourosopy	1		
Thal Type	Normal		

Figure 4.4 Clinical Report Generation

The Clinical Report Generation module automatically creates a detailed medical assessment report based on the patient’s prediction results. The generated report includes patient information, clinical parameters, risk percentage, model diagnostics, and final prediction outcomes.

PDF Report Generation :

The system generates downloadable clinical reports in PDF format for easy sharing and future medical reference. The report is structured similarly to professional healthcare diagnostic documents and contains detailed information regarding patient health analysis and Machine Learning predictions.

Report Contents and Analysis :

The generated report contains:

- Patient profile details
- Medical input parameters
- Prediction confidence score
- Model-wise prediction results
- Final ensemble prediction outcome

The report generation feature improves usability by allowing users and healthcare professionals to maintain prediction records for future comparison and medical consultation.

User-Friendly Design :

The report interface uses a clean and professional design with clear section headings, structured tables, and readable formatting. Important results such as high-risk predictions are highlighted prominently to ensure quick interpretation and effective healthcare decision-making.

CHAPTER 5
EXPERIMENTAL SETUP AND RESULTS

5.1 EXPERIMENTAL SETUP

Hardware Requirements:

Component	**Specification**
Processor	Intel Core i7 or equivalent
RAM	8 GB or higher
Storage	256 GB SSD or higher
Graphics card	Nvidia RTX
3040	

Software Requirements:

Component	**Specification**
Operating System	Windows 11
IDE / Code Editor	Visual Studio Code
Database	MongoDB
Browser	Google Chrome
Deployment Platform	Localhost

Programming Environment:

Category	**Tools / Technologies Used**
Frontend	HTML5, CSS3, JavaScript
Backend	Python Flask
Database Query	SQLite3
Machine Learning Library	Scikit-learn
Report Generation	PDFKit / ReportLab

5.2 RESULTS

The experimental results showed that the Heart Disease Prediction System efficiently analyzed patient clinical data and generated accurate heart disease risk predictions using ensemble Machine Learning techniques. The integrated Machine Learning models, including Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Decision Tree, and Random Forest, achieved high prediction accuracy and improved reliability through majority voting. The ensemble approach enhanced prediction consistency and reduced the limitations of individual models.

System testing confirmed that the application was responsive, accurate, and user-friendly, providing clear prediction results, confidence scores, and detailed clinical reports. The automated report generation and prediction visualization features improved accessibility and helped users better understand their health assessment results. Overall, the system successfully achieved its objective of providing an intelligent, reliable, and efficient heart disease prediction platform for early healthcare support and decision-making.

CHAPTER 6
CONCLUSION AND FUTURE WORKS

The proposed Heart Disease Prediction System provides an intelligent and efficient solution for predicting cardiovascular disease risk using Machine Learning techniques. By integrating multiple classification algorithms and ensemble learning methods, the system improves prediction accuracy, reliability, and robustness compared to traditional single-model approaches. The Medical Input and Data Processing Module ensures accurate collection and preprocessing of patient clinical information, while the Machine Learning Prediction Module analyzes the data to generate reliable heart disease risk assessments.

The ensemble voting mechanism combines multiple Machine Learning algorithms to generate accurate and reliable heart disease predictions. The system's user-friendly dashboard, prediction visualization, and automated report generation improve accessibility for both patients and healthcare professionals while supporting early disease detection through intelligent healthcare technology.

The Heart Disease Prediction System offers significant opportunities for future enhancement and expansion. One major improvement would be the integration of deep learning models such as Artificial Neural Networks (ANN) and Long Short-Term Memory (LSTM) networks to further improve prediction accuracy and handle larger medical datasets more effectively. The system could also incorporate real-time health monitoring through wearable IoT devices such as smartwatches and heart-rate sensors, enabling continuous patient monitoring and dynamic risk analysis.

Another promising enhancement is the integration of Explainable Artificial Intelligence (XAI) techniques to provide detailed explanations for prediction outcomes, helping doctors and patients better understand the factors influencing heart disease risk. The application could also be extended into a mobile healthcare platform that allows users to perform health assessments, receive emergency alerts, and monitor their medical history remotely.

Future enhancements to the system may include advanced healthcare analytics, personalized treatment recommendations, hospital integration, and cloud-based patient data management. Using larger and more diverse medical datasets can further improve prediction accuracy and reliability, transforming the system into a more intelligent and efficient AI-powered healthcare assistant.

CHAPTER 7
REFERENCES

1. D. Dua and C. Graff, “UCI Machine Learning Repository: Heart Disease Dataset,” University of California, Irvine, School of Information and Computer Sciences, 2023.
2. S. Rajesh and T. Dharun, “Heart Disease Prediction System Using Machine Learning,” International Journal of Engineering Research and Technology (IJERT), vol. 9, no. 4, pp. 659–662, 2020.
3. K. Anderson, P. Lee, and M. Roberts, “Machine Learning Approaches for Cardiovascular Disease Prediction,” IEEE Access, vol. 10, pp. 45678–45689, 2022.
4. A. Sharma and R. Kumar, “Heart Disease Prediction Using Logistic Regression and Random Forest,” International Journal of Advanced Computer Science and Applications (IJACSA), vol. 12, no. 5, pp. 211–217, 2021.
5. J. Patel and S. Shah, “Prediction of Heart Disease Using K-Nearest Neighbor and Support Vector Machine Algorithms,” Procedia Computer Science, vol. 167, pp. 1150–1159, 2020.
6. M. Singh, R. Verma, and P. Gupta, “Ensemble Learning Techniques for Heart Disease Prediction,” International Conference on Computational Intelligence and Data Science (ICCIDS), pp. 342–349, 2022.
7. F. Chollet, “Deep Learning and Healthcare Analytics,” Springer International Publishing, pp. 188–205, 2021.

8. A. Brown and L. Wilson, “Explainable Artificial Intelligence in Healthcare Systems,” IEEE Transactions on Artificial Intelligence, vol. 3, no. 4, pp. 420–431, 2023.
9. Scikit-learn Developers, “Scikit-learn Machine Learning Library Documentation,” Scikit-learn Official Documentation, 2024.
10. Flask Documentation Team, “Flask Web Framework Documentation,” Pallets Projects Documentation, 2024.
11. P. Kumar and S. Mehta, “Heart Disease Prediction Using Decision Tree and Random Forest Algorithms,” International Journal of Computer Applications, vol. 182, no. 44, pp. 15–21, 2021.
12. R. Chandra and V. Sharma, “Medical Data Classification Using Machine Learning Algorithms,” Journal of Biomedical Informatics, vol. 118, pp. 103–114, 2022.
13. NumPy Developers, “NumPy Reference Guide,” NumPy Official Documentation, 2024.
14. Pandas Development Team, “Pandas Data Analysis Library Documentation,” Pandas Official Documentation, 2024.
15. T. Johnson and A. Smith, “Clinical Decision Support Systems Using Artificial Intelligence,” International Journal of Healthcare Informatics Research, vol. 7, no. 2, pp. 98–110, 2023.